

Nidhi Parab

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EDUCATION

University at Buffalo

Masters of Science in Computer Science and Engineering

Buffalo, NY

2024 – 2026

University of Mumbai

Bachelor of Engineering (9.5/10 CGPA)

Mumbai, India

2020 – 2024

EXPERIENCE

Undergraduate Research Student

Indian Institute of Technology (IIT), Bombay

June 2023 – June 2024

Mumbai, India

- Hyperspectral Images are 3D images consisting of multiple wavelengths, providing significantly more information than standard RGB images. HSI Classification faces challenges due to limited datasets, often leading to overfitting and poor generalization to new data.
- To overcome these challenges, we enhanced a Few Shot Learning method by integrating Self-Attention and Temperature Scaling into the model. This innovative approach aimed to improve the model's accuracy and robustness with minimal training samples.
- I was responsible for designing the Self-Attention network architecture, conducting ablation studies, and performing result analysis. Our experiments were conducted on five leading Hyperspectral Image datasets: Indian Pines, Salinas, Kennedy Space Center, Botswana, and Pavia University
- We achieved an impressive accuracy of 99.86%, surpassing the state-of-the-art (SOTA) methods by saving the total time required to classify the HSI patches.

Technical Coach

CodePath

May – Aug. 2023

San Francisco, CA

- Coach for a 10-week online course on 'Advanced Technical Interview Preparation,' mentoring a cohort of 5 students to excel in their technical interviews. Among which 2 secured internship in Dell and Microsoft.
- Taught and provided guidance on essential programming concepts including linked lists, arrays, trees, dynamic programming, backtracking, and greedy algorithms.

SDE Intern

Indian Development Foundation

Dec. 2022 – Mar. 2023

Mumbai, India

- Developed an all-in-one platform called 'Balgurukul' for 300 non-profit schools.
- The centralized interface facilitates teachers in grading students, while also enabling organizational stakeholders to manage personal data of teacher volunteers across the nation.
- Developed a solution using Nextjs and MySQL to facilitate efficient oversight of staff, students, and grading processes, ultimately optimizing operational workflows and fostering an environment conducive to educational excellence.

PROJECTS

GitHub Issue Finder CLI | *Rust*

Oct. 2024

- Built a command-line tool to find and display "good-first-issue" GitHub issues using the GitHub API.
- Utilized the GitHub Search Issues API to fetch and filter issues labeled "good-first-issue" from open repositories.
- Implemented asynchronous programming with *Tokio* for efficient handling of multiple API requests.
- Applied *request* for HTTP requests and *serde* for JSON parsing, ensuring robust error handling.

RustMust | *Rust*

Jun. - Aug. 2024

- Developed a collection of Rust projects focusing on essential daily functions.
- Implemented file compression and decompression utilities to efficiently reduce and restore file sizes.
- Created a tool for bundling multiple files into a single zip folder.
- Built functions to create, read, and process JSON and CSV data.

Yogi - The Yoga Instructor | *Flask, OpenCV, Media Pipe, PHP, HTML*

Sep. 2022 – Apr. 2023

- Developed and implemented an advanced artificial intelligence (AI) system that guides its users through personalized yoga sessions, providing real-time feedback and instruction indicating the accuracy of posture performed.

Hand-2-Hand | *React Native, Google Firestore*

Feb. – May 2022

- Developed a full-stack mobile application using React Native.
- Innovative e-commerce application facilitating the sustainable exchange of study materials between college seniors and juniors, addressing the challenges posed by the pandemic and promoting cost-effective learning solutions.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Rust

Frameworks: React, Node.js, Flask

Developer Tools: Git, VS Code, PyCharm, IntelliJ, Eclipse

Libraries: Pandas, OpenCV, Media Pipe NumPy, Matplotlib

PUBLICATION

Personalized AI Based Yoga-Pose Recommendation System Using Game Theory: Achieved the ‘Best Paper Award’ in the Artificial Intelligence Track among 120 participants at the 3rd International Conference on Advanced Computing Technologies and Applications (October 2023) organized by IEEE Bombay Section, Mumbai.

LEADERSHIP

Microsoft Student Learn Ambassador, 2023-24. Led 2 workshops as a key advocate for Microsoft technologies, fostering a community of learners.

General Secretary at the Computer Society of India (CSI) 2023-24. Overseeing tasks for all the events and coordinating the work of 32 committee members .

AWARDS AND CERTIFICATIONS

- Selected among the top 40 in India among 300 applicants, for ‘Network Science’ at the Association for Computing Machinery (ACM) Winter School, Ahmedabad University, Gujarat, December 2023
- Featured speaker on ‘Hyperspectral Image Classification’ at annual distributed tech conference, DevFest Mumbai 2023, attended by over 100 participants held at WeWork, Mumbai, December 2023.
- Silver Elite Certificate in Java by National Programme on Technology Enhanced Learning (NPTEL) and Indian Institute of Technology (IIT) Kharagpur